



OPEN Improved clinical pregnancy rates in natural frozen-thawed embryo transfer cycles with machine learning ovulation prediction: insights from a retrospective cohort study

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This study aims to develop physician support software for determining ovulation time and assess its impact on pregnancy outcomes in natural cycle frozen embryo transfers (NC-FET). To develop, assess, and validate an ovulation prediction model, three datasets were used: REI Ovulation Determination dataset (500 cycles) split into training (309), validation (90), and test (101) sets; the Documented Ovulation dataset (101 cycles) with confirmed ovulation (documented follicular rupture and LH surge); and the Clinical Pregnancy Rates dataset (515 NC-FET cycles), categorized into “Matched” and “Mismatched” based on alignment with the model’s ovulation determination. Pregnancy outcomes were compared between the groups. The ovulation prediction model exhibited 93.85% and 92.89% matching rates with the REI Ovulation Determination and Documented Ovulation datasets, respectively. In the Clinical Pregnancy Rates dataset, the Matched group (282 cycles) showed significantly higher clinical pregnancy rates than the Mismatched group (34.6% vs. 25.9%, $p = 0.04$) and similar results for patients under 37 (41.1% vs. 30.7%, $p = 0.04$). Logistic regression indicated lower pregnancy rates in Mismatched cases (odds ratio 0.67 for the general population, 0.63 for patients under 37). In conclusion, we introduce a highly accurate AI ovulation prediction model. Treatment cycles aligning with the model’s recommendations had significantly increased clinical pregnancy rates.

Key message

This study introduces the first AI model designed for predicting ovulation, proving its high accuracy. Treatment cycles aligning with the model’s recommendations had significant 8.7% increase in clinical pregnancy rates, emphasizing the potential of AI in optimizing NC-FET outcomes.

Keywords Ovulation, Machine learning, Frozen embryo transfer, Natural cycle, Prediction

The recommended approach for frozen embryo transfer (FET) is the utilization of a natural cycle, the NC-FET, which relies on ovulation detection¹. NC-FET presents several advantages compared to hormone replacement therapy (HRT) cycles. It eliminates the need for hormone intake, tends to be more cost-effective, and has robust evidence indicating a protective effect against long-term obstetric complications^{2,3}. These factors contribute to making natural NC-FET a more appealing choice. Nonetheless, there are drawbacks to NC-FET. Success depends on precisely timing ovulation and aligning embryo development with the endometrial stage, requiring more clinic visits^{1,4}.

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Earlier research has investigated the precise methods for ascertaining the timing of ovulation, considering factors such as hormone absolute levels or variations and the disappearance of follicles in ultrasound examinations^{5–8}. The determination of ovulation timing involves a combination of data, including the key hormones in the ovulation process (LH, estrogen, and progesterone), as well as ultrasound monitoring of follicle size and the timing of rupture⁹. These components serve as tools for clinicians when deciding on ovulation timing and establishing the date for embryo transfer¹⁰. Throughout the follow-up process, physicians must make a series of critical decisions that significantly impact the cycle's outcome.

In recent times, artificial intelligence (AI) models have been integrated into various domains, with medicine being no exception. Within the realm of fertility treatment, this technology may enhance treatment outcomes, ensuring consistency in global data analysis, and refining large-scale data analysis. Numerous groups are investigating machine learning technologies in the fertility field, addressing tasks ranging from predicting outcomes in in vitro fertilization¹¹, to ovarian stimulation^{12,13}, intrauterine insemination¹⁴, sperm selection¹⁵, optimal trigger day^{16,17}, embryo selection¹⁸ and euploidy¹⁹. Furthermore, in recent years, there has been a surge in the global utilization of fertility treatments. This escalation poses a significant challenge for healthcare professionals, leading to both increased workload for doctors and delays in treatment for patients. The most significant increase has been observed in frozen embryo transfers²⁰ and NC-FET as the preferred method. Recognizing the paramount importance of precisely determining ovulation time, especially in the context of frozen embryo transfer treatments, and in an effort to streamline and potentially enhance the work of healthcare providers, a decision-support machine learning software for accurately determining ovulation has been developed. In this study, we aimed to test the proposed model and its accuracy in determining the correct ovulation timing and accordingly the embryo transfer timing in NC-FET.

Results

Model calibration

The model was calibrated as part of the training process and reached an expected calibration error (ECE) of 2.99 on the test set of the REI Ovulation Determination Dataset.

Feature importance

The top ten important features, used by the model to determine ovulation day, are presented in Fig. 1. Features related to progesterone, LH, estrogen, the ratio between estrogen and progesterone, and leading follicle size on the instance's last test day are the five most important ones. Interestingly, progesterone on the second day was the most important feature. The other features used by the model were: endometrial thickness, estrogen value on the instance's first test day, and features related to LH, progesterone, and estrogen relative changes between the instance's first and second test day.

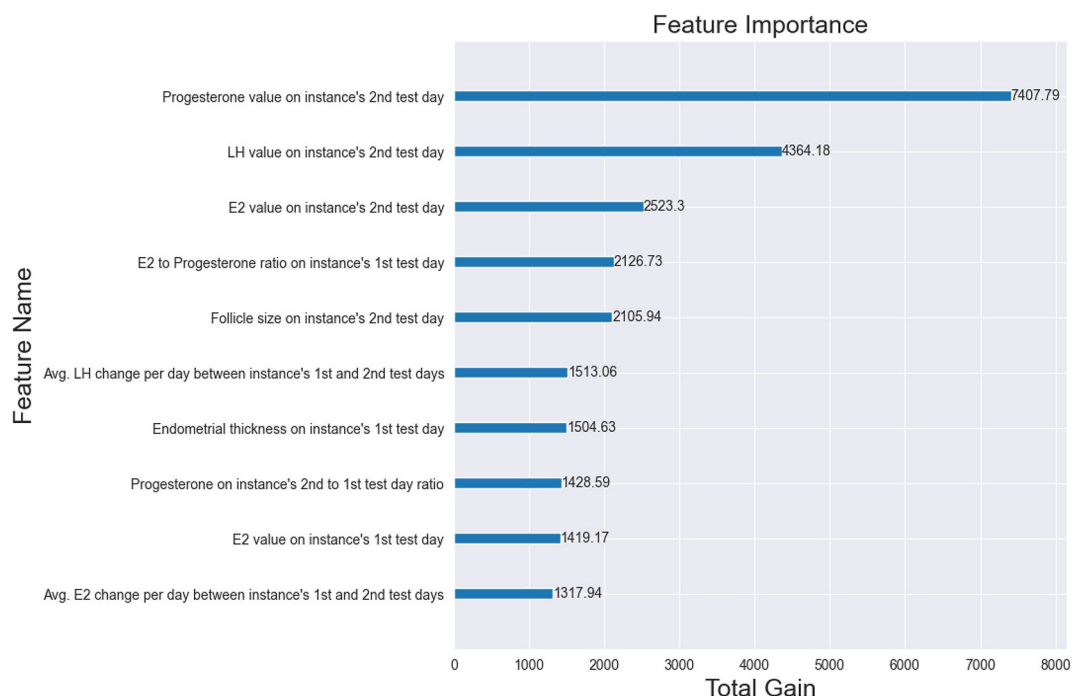


Fig. 1. Summary of the model's top ten most important features by total gain of each feature. Progesterone and LH on the instance's second test day were the most important features.

Ovulation prediction model accuracy

The model's performance on each test set is represented as a confusion matrix (Fig. 2A and B). The model achieved a total accuracy of 93.85% in predicting the ovulation day in the REI Ovulation Determination Dataset's test set, and 92.89% in the Documented Ovulation Dataset. The model had similar performance on both datasets, showing high precision on days ≤ -2 , -1 , and 0 while performing less well on the limited number of instances on days 1 and ≥ 2 .

Clinical pregnancy rates

Table 1 illustrates the demographic and cycle characteristics of the two study groups, Matched and Mismatched. No significant differences were identified between the two groups in parameters that could potentially impact pregnancy rates, such as the patient's age at the time of embryo freezing or transfer, BMI, number of previous oocyte retrievals, and embryo quality.

The sole parameter where statistical significance emerged was in the measurement of endometrial width. The endometrium width on the last test day was 9.9 ± 1.8 mm in the Matched group compared to 9.6 ± 1.6 mm in the Mismatched group. Due to the minimal difference in endometrium thickness and the inclusion of cases where the endometrium width was equal to or surpassed 7 mm, we believe that this variation has no clinical significance and does not have any influence on the outcomes of the study.

A comparison of cycle outcomes including pregnancy rates is presented in Table 2. Notably, the Matched NC-FET group exhibited a significantly higher clinical pregnancy rate of 34.6%, compared to the Mismatched group with a rate of only 25.9%.

In the subsequent analysis, we performed two multivariate logistic regressions for clinical pregnancy. The first analysis included all age groups, while the second one included only patients younger than 37 years. For the all-ages-group analysis, we show a notable decrease in pregnancy rates when there was a lack of agreement with the attending physician (Table 3), indicating that instances of mismatch could lead to a 33% reduction in pregnancy rates (OR 0.67, 95% CI 0.54–0.99). As expected, the patient's age at the time of egg retrieval emerged as a significant factor influencing the clinical pregnancy rate. All other parameters were non-significant. A similar decrease of 37% in pregnancy rate for the mismatch group was found in the analysis of the younger patients group. However, the patient age influence was not observed (Table 4).

The Mismatched group can be subdivided into two categories: Early and Late Mismatched. The Early Mismatched category comprises cases where the embryo was transferred earlier than the suggested date based on the model-determined ovulation dates where the embryo was transferred later than the suggested date according to the model. A multivariate logistic regression including these groups indicated a significant decrease in the likelihood of conception in the Late Mismatched group compared to the Matched group. Early transfers also exhibited a decrease in the probability of achieving pregnancy, however it was not statistically significant. Detailed results are provided in Table 3 in the supplementary data.

Discussion

Artificial intelligence technology offers numerous opportunities for advancements in medicine overall, with a specific emphasis on fertility medicine. In this study, we present, for the first time, an accurate novel AI model for determining ovulation. The model demonstrated a 93% accuracy across two distinct validation methods. A recommended transfer date was calculated using the model's ovulation determination, and in cases where there was agreement between the physician and the model on transfer date in NC-FET cycles, there was a significant increase of 8.7% in clinical pregnancy rates.

Despite being a routine practice worldwide, determining ovulation remains a challenge. Historically, the disappearance of the leading follicle marked ovulation, and many studies have hinged on this criterion^{10,21,22} due to its consistent nature. However, Ecohard and colleagues, in a study tracking 271 ovulation cycles, reported only 84.3% sensitivity and 89.2% specificity for follicular rupture as an indicator of ovulation²³. An alternative method to detect ovulation is the onset of the LH surge. However, this method has its own inherent limitations, particularly due to the absence of consensus regarding the precise definition of the LH surge. The definitions found in the literature vary, ranging from a relative change in LH levels compared to the previous measurement²⁴ to the use of absolute cutoffs LH of 10 IU/l²⁵, 15 IU/l²⁶, 17 IU/l⁸, or 20 IU/l or more²⁷. In a previous study⁹, we demonstrated that relying solely on relative changes or absolute values of LH to predict ovulation resulted in unsatisfactory performance. Consequently, we conclude that in a clinical setting, a comprehensive approach combining LH surge determination with estrogen and progesterone hormonal assessments and ultrasound is essential for accurate ovulation determination.

Subsequently, to the above-mentioned challenge of ovulation determination, the model's ability to predict ovulation was subjected to testing in three distinct contexts: cycles where ovulation determination was carried out by fertility REIs, cycles with documented ovulation (determined by both LH surge and follicular rupture), and a third test assessing the model's impact on clinical outcomes using a recommended transfer date of five days after model's ovulation determination compared to the true transfer date selected by the attending physician.

Interestingly, the concordance between the model prediction and the two distinct ovulation determination validation methods showed very similar results, with an accuracy of 93% for both sets, strengthening the confidence in the model's accuracy. Moreover, our findings revealed a significantly higher pregnancy rate in the group where there was concordance on the day of transfer between the model and the attending physician in the Clinical Outcomes Dataset.

Determining the date of embryo transfer in a natural cycle is a complex question and there are several approaches to this matter. In this study, the recommendation for blastocyst transfer by the machine was scheduled 5 days after ovulation (ovulation + 5), aligning with the common practice of LH surge + 6 days^{1,3,28}. However,



Fig. 2. A. Model performance on the REI ovulation determination test set displayed as a confusion matrix. B. Model performance on the Documented Ovulation Dataset displayed as a confusion matrix. Columns represent instances' actual classes, relative to the instance's latest test day. Rows represent the model-predicted classes. Number and percentage of occurrences for actual vs. prediction pairs are presented in each cell. Percentages were normalized cumulative sum of 100%. The bottom row and right column highlight the accuracy in green and the errors in red, respectively. The number of occurrences is positioned at the top of each cell. The bottom-right cell signifies the overall model accuracy.

	Matched group n = 283	Mismatched group n = 232	P-value
Patient age (at oocyte retrieval) / Mean $\pm$ SD (25–75 Percentiles)	34.3 $\pm$ 5.2 (30–39)	34.6 $\pm$ 5.3 (31–39)	0.51
Patient age (at transfer) / Mean $\pm$ SD (25–75 Percentiles)	34.9 $\pm$ 5.2 (31–39)	35.1 $\pm$ 5.2 (31–39.7)	0.69
Patient BMI / Mean $\pm$ SD (25–75 Percentiles)	23.9 $\pm$ 4.7 (20.4–26.4)	24.0 $\pm$ 5.4 (20.1–26.2)	0.66
Previous retrievals No. / Mean $\pm$ SD (25–75 Percentiles)	1.9 $\pm$ 1.7 (1–2)	1.7 $\pm$ 1.4 (1–2)	0.18
No. of blood tests Mean $\pm$ SD (range) (25–75 Percentiles)	3.6 $\pm$ 1.4 (3–5)	3.6 $\pm$ 1.3 (3–5)	0.59
No. of US tests Mean $\pm$ SD (25–75 Percentiles)	3.6 $\pm$ 1.4 (3–5)	3.6 $\pm$ 1.3 (3–5)	0.83
Endometrial thickness mm/ Mean $\pm$ SD (25–75 Percentiles)	9.9 $\pm$ 1.8 (8.6–11)	9.6 $\pm$ 1.6 (8.5–10)	0.045
Quality of transferred embryos (%) A B	42.8 57.8	37.9 62.1	0.27

Table 1. Demographic and cycle characteristics data of the Matched and the mismatched study groups.

	Matched n = 283	Mismatched n = 232	P-value
Transfer day according to physician/Mean $\pm$ SD (25–75 Percentiles)	21.0 $\pm$ 3.4 (19–23)	20.0 $\pm$ 3.7 (17–22)	< 0.01
Transfer day according to model/Mean $\pm$ SD (25–75 Percentiles)	21.0 $\pm$ 3.4 (19–23)	20.5 $\pm$ 3.6 (18–23)	0.07
Pregnancy rate (%)	37.8	28.9	0.033
Clinical pregnancy rate (%)	34.6	25.9	0.032
Subgroup analysis - Age $\leq$ 37y	Matched n = 190	Mismatched n = 152	
Pregnancy rate age $\leq$ 37y (%)	44.8	34.7	0.058
Clinical pregnancy rate age $\leq$ 37y (%)	41.1	30.7	0.046

Table 2. Matched group and mismatched group cycle results.

Parameter	Odds Ratio	Confidence interval 95%	P-value
Patient BMI	0.99	0.95–1.03	0.74
Endometrial thickness	1.03	0.92–1.14	0.63
Embryo quality*	1.08	0.73–1.59	0.71
No. of previous retrievals	1.01	0.9–1.15	0.70
Patient age at oocyte retrieval	0.95	0.91–0.98	0.005
Physician decision Mismatched to model	0.67	0.45–0.99	0.043

Table 3. Odds ratio for clinical pregnancy rate - all age groups. A multivariate logistic regression.

Parameter	Odds Ratio	Confidence interval 95%	P-value
Patient BMI	0.99	0.95–1.04	0.87
Endometrial thickness	1.01	0.90–1.15	0.80
Embryo quality*	1.00	0.64–1.59	0.99
No. of previous retrievals	0.88	0.70–1.11	0.29
Patient age at oocyte retrieval	1.03	0.97–1.10	0.28
Physician decision Mismatched to model	0.63	0.40–0.99	0.046

Table 4. Odds ratio for clinical pregnancy rate in patients younger than 37 years. A multivariate logistic regression. * Odds ratio for transfer of embryos grade A quality vs. grade B embryos.

there are diverse strategies for scheduling frozen embryo transfer post-LH surge. Irani and colleagues⁸ proposed a schedule of 5 days following the onset of the LH surge. Although the window of implantation might allow for some variation, the importance of accurate determination of ovulation cannot be over-emphasized as shown by the decrease in pregnancy rate in this study. Interestingly, a significant difference was observed between the Matched and Late Mismatch groups. Yet, the determination of the optimal timing for frozen embryo transfer, whether rooted in follicular rupture, LH surge, or progesterone levels, falls outside the scope of this article.

Several factors may contribute to the variance between the physician's decision on the transfer date and the date suggested by the model. These factors include potential inaccuracies in determining the ovulation date, challenges in coordinating a convenient transfer date for both the doctor and the patient, or adherence to various protocols related to the optimal day of transfer (the model's determination was compared with the true transfer date of each cycle scheduled by the attending physician according to his scheduling strategy). The utilization of the model for ovulation determination and subsequent transfer date scheduling has the potential to improve the chances of achieving pregnancy.

The model used multiple features to predict ovulation (Fig. 1). The most influential features affecting the model's performance were progesterone, LH, and estrogen on the instance's last test day. This finding demonstrates that using a combination of all 3 hormones is beneficial to accurately predicting ovulation, and the use of any single-hormone-methods may reduce accuracy. The strength of utilizing artificial intelligence lies in its ability to perform complex analysis on countless combinations of multiple indicators that when assessed alone, might exhibit limited effectiveness. In our clinical setting, blood tests and ultrasounds are typically performed every 2–3 days, which balances patient convenience, cost, and clinical requirements, but does not provide continuous daily monitoring. This highlights the strength of our machine learning approach, as it can still deliver accurate predictions even when frequent monitoring is not feasible, such as in cases where the follicle disappearance is not observed directly. The algorithm's ability to predict ovulation under these conditions underscores its practical utility in real-world clinical scenarios, particularly where continuous daily assessment is impractical.

Within natural cycles, there are two main approaches: the conventional natural cycle and the modified natural cycle. In this study, only cycles in which the NC-FET protocol was chosen by the attending physician were included. Although several studies have investigated whether the modified NC-FET protocol is superior to the conventional NC-FET^{29–32}, this issue is beyond the scope of the current article. Nevertheless, it raises intriguing questions about the role of artificial intelligence (AI) in optimizing modified natural cycles. For example, could AI improve outcomes by better timing the trigger? We believe that further studies addressing this topic will likely be available in the future.

This model, in addition to being used to assist in ovulation determination in NC-FET, can be used to develop more advanced algorithms for various clinical scenarios such as NC-FET, IUI, or timed intercourse treatment management. The model can also efficiently label the ovulation day in a vast number of cycles that can be used to train more precise future models, a methodology known as teacher-student in the machine learning literature. Youngster et al.¹⁴ used such a method to train a model capable of predicting ovulation up to 6 days in advance, and effectively and automatically managing IUI cycles. Using this method to automatically manage NC-FET or modified-FET cycles is a prospect for future research.

The REI opinion and Documented Ovulation Datasets both have inherent limitations due to the non-random nature of the Documented Ovulation Dataset, and reliance on the REI-opinion of the REI Ovulation Determination Dataset. However, the model's consistent performance on both datasets suggests its predictions are accurate and reliable. The pregnancy outcomes analysis of this study is limited by its retrospective nature. To overcome these limitations, future prospective randomized control studies as well as external validation in different medical centers are needed.

In conclusion, in this study, we present an innovative and accurate AI model designed for ovulation detection. This model can be used to develop more advanced algorithms for various clinical scenarios including NC-FET, IUI, and timed intercourse treatment management. The study outlines the promising integration of artificial intelligence techniques into assisted reproductive technology treatments, emphasizing its potential to optimize treatment workflow and clinical outcomes.

Materials and methods

This research introduces an innovative machine-learning model designed to accurately determine the day of ovulation in NC-FET cycles.

Study Design and patients

This retrospective single-center cohort study was approved by the Herzliya Medical Center Institutional Review Board (Study no. HMC-0008–21) and all methods were carried out in accordance with relevant guidelines and regulations. Due to the retrospective nature of the study, Herzliya Medical Center Institutional Review Board waived the need of obtaining informed consent. The center offers laboratory and surgical services to over fifty fertility reproductive specialists. Each of these physicians, consider various factors when selecting a treatment protocol, including personal preference, patient desires, and other considerations. In this study, only cycles where the NC-FET protocol was chosen by the physician were included.

In this study, 1,116 NC-FET performed between August 2018 and July 2023 were used and divided into three datasets. Patient information was de-identified before analysis.

Data Preparation

The data obtained for each cycle consisted of baseline characteristics, including the patient's age at the time of embryo freezing, age at the time of embryo transfer, body mass index (BMI), and cycle characteristics including

the leading follicle size, endometrial thickness, and blood test results for estradiol, progesterone, and LH levels. Table 1 in the supplementary material and methods summarizes patients’ demographics and cycle characteristics.

Data cleaning

In this study, only true NC-FET cycles were utilized. Stimulated, HRT, and modified NC-FET were excluded. No imputation was performed to handle missing values, the model was designed to learn from their occurrence and give its prediction despite them being missing.

Dataset creation

This study includes three distinct and non-overlapping datasets. A total of 1,476 NC-FET cycles, each including at least two blood tests and an ultrasound, conducted between August 2018 and July 2021, were used to form the first two datasets: the REI Ovulation Determination Dataset and the Documented Ovulation Dataset.

REI ovulation determination dataset

The REI Determination Dataset included 500 randomly selected cycles from the initial 1476 cycles. This group was used to train, validate, and test the ovulation determination model. Determination of ovulation days (Ground Truth) in this dataset was carried out by a majority decision among three independent reproductive endocrinologists, each with an average of 15 years of extensive experience. The cycles were split into 309 training cycles, 90 validation cycles, and 101 testing cycles (Table 5).

Documented Ovulation dataset

The Documented Ovulation Dataset included 101 selected cycles and was used to evaluate the ovulation determination model accuracy in predicting ovulation without relying on REI determination. Inclusion criteria were: two consecutive days of blood and ultrasound tests, documented follicular rupture of a single primary follicle measuring ≥ 17 mm, and documented LH surge, defined as LH ≥ 17 IU on the initial day, with a reduced value on the subsequent day. Figure 3A summarizes the creation of these datasets.

Clinical pregnancy Rates dataset

The third group was constructed from 2,759 NC-FET cycles that occurred between August 2021 and August 2023. After applying the strict inclusion criteria outlined in Fig. 3B, this dataset was refined to form the 515 cycles of the Clinical Pregnancy Rates Dataset. Unlike the previous two datasets, these cycles were used to assess the impact of transferring embryos in accordance with the prediction of the trained ovulation determination model on pregnancy outcomes. All included cycles had ovulation determined by the model, and a recommended transfer date 5 days after ovulation (ovulation + 5). Next the model’s determination was compared with the true transfer date of each cycle scheduled by the attending physician according to his scheduling strategy. The dataset was then divided into two groups: Matched cycles in which the transfer occurred according to the model’s recommendation and Mismatched cycles in which transfer did not occur according to the model’s recommendation. The pregnancy rates and confounding factors were compared between the two groups. It is important to note that the clinical pregnancy rates dataset was separated to a later time period for two main reasons: 1) To isolate it from being affected by the removal of cycles that were selected to be used in the model development and testing datasets. 2) High availability of pregnancy data during that time period.

Figure 3: (A) Flowchart of patient selection in the REI Ovulation Determination Dataset and Documented Ovulation Dataset. (B) Flowchart of patient selection in the Clinical Pregnancy Rates Dataset. Out of 2,759 NC-FET cycles, 515 (283 Matched and 232 Mismatched) cycles were used after filtering.

* Blastocyst classification was based on the Istanbul Consensus Workshop^{29,30} and was adjusted to the local laboratory standards, resulting in three quality grading groups. Summarizing table is in the supplementary data section (Supplemental Table 4).

Instances Creation

To train the model, multiple instances were created from each cycle in the REI Ovulation Determination Dataset’s training and validation sets, where each instance included a different combination of test days from that cycle. The instances included all single test days and any combination of two test days that are at most four days apart. Training the model with additional test days or test days further apart did not improve its performance.

To evaluate the model’s performance, instances were created from cycles in the REI Ovulation Determination Dataset’s test set and the Documented Ovulation Dataset. For this purpose, all combinations of test day pairs, with at most four days apart, were used. Additionally, if a test day was not included as part of a pair, that day was tested individually.

The target class for each instance is determined by calculating the difference between the instance’s last test day and the ovulation day. The five ovulation classes are defined as two or more days before (≤ -2), a day before

Dataset	Cycles				Instances			
	Total	Train	Validation	Test	Total	Train	Validation	Test
REI Ovulation Determination	500	309	90	101	3,181	2,155	603	423
Documented Ovulation	101			101	591			591

Table 5. Summary of the number of cycles and instances per dataset.

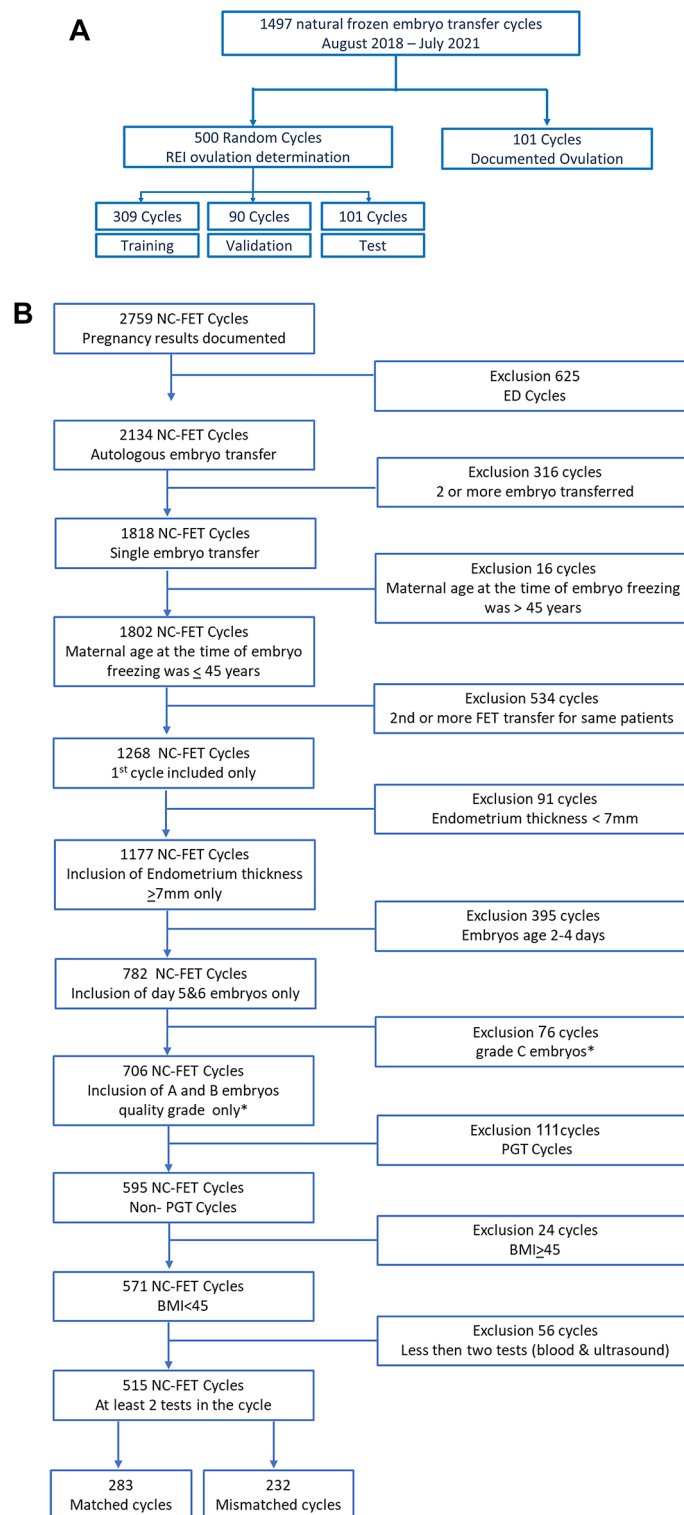


Fig. 3. A. Flowchart of patient selection in the REI Ovulation Determination Dataset and Documented Ovulation Dataset. B. Flowchart of patient selection in the Clinical Pregnancy Rates Dataset. Out of 2,759 NC-FET cycles, 515 (283 Matched and 232 Mismatched) cycles were used after filtering.

(−1), the same day as (0), a day after (+1), or two or more days after (≥ 2) the ovulation day. For example, Table 1 in supplementary material and methods shows an example of a cycle with ovulation on the 13th day, while the blood tests were performed on the 3rd, 10th, 12th, and 14th days. As can be seen from the example, seven possible instances can be generated.

From the 500 cycles in the REI Ovulation Determination Dataset, 3,181 instances were generated. These instances were randomly split by the patient identifier, into the train (~67%), validation (~20%), and test (~13%) sets, while maintaining the same class distribution between the sets. From the 101 cycles in the Documented Ovulation Dataset, 591 instances were generated. Table 5 summarizes the dataset groups and their characteristics.

Model Design

The model utilizes data obtained from up to 2 days of blood and ultrasound tests. Based on instances that were extracted from the REI Ovulation Determination Dataset, described in the [data preparation](#) section, a machine learning model was trained. The model was trained on the train set to be able to classify, given an instance, the prediction of the ovulation class (≤ -2 , -1 , 0 , $+1$, $\geq +2$). The features included information related to each instance consisting of baseline characteristics, and features extracted from the cycle characteristics including ultrasound tests and blood test results.

XGBoost classifier, a machine learning model, was selected due to the combination of its high performance and its powerful handling of missing values. The model features were chosen by using the Forward Sequential Feature Selection (Forward-SFS), which resulted in a subset of 20 features. See Supplementary materials and methods for the selected features for the detailed list of features used for the model learning. Multi-class calibration using one versus rest methodology was performed as a part of the models' development. The grid search cross-validation method was applied and evaluated on the validation set to optimize the model hyperparameters, such as learning rate and maximal depth. After the hyperparameters optimization, the final model was trained on the training and validation sets. During the prediction stage, the ovulation class with the greatest likelihood, per instance, was selected as the model's final prediction.

Statistical analysis

The distribution of continuous variables was evaluated using histogram and Q-Q plot. The distribution of continuous variables was reported as mean and interquartile range. Continuous variables were compared between Matched and Mismatched using the Mann-Whitney test or the Kruskal-Wallis test (the Dunn test was used for post hoc analysis). Categorical variables were summarized as frequency and percentages. Categorical variables were compared using the Chi-Square test.

Multivariable logistic regression was used to study the association between Matched and Mismatched and clinical pregnancy while controlling for potential confounders.

All statistical tests were two-sided and $p < 0.05$ was considered a statistical significance. SPSS software was used for statistical analysis (IBM Corp. Released 2021. IBM SPSS Statistics for Windows, Version 28.0. Armonk, NY: IBM Corp).

The analysis was conducted across all age groups, including a subgroup of patients under 37 years of age. This threshold was selected as it represents the upper limit for younger age groups according to the classifications defined by the Society for Assisted Reproductive Technology (SART)³¹ and the National Institute for Health and Care Excellence (NICE) guidelines³².

Data availability

Data is provided within the manuscript or supplementary information files. The data underlying this article will be shared on reasonable request to the corresponding author.

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Declarations

Competing interests

Competing Interests: Authors: Ettie Maman, Rohi Hourvitz, Almog Luz, Micha Baum, and Ariel Hourvitz are shareholders and board members of FertilAI LTD. Authors Eden Moran, Nevo Itzhak and Shachar Reuveny are FertilAI employees. Author Michal Youngster has nothing to declare.

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Additional information

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